# Derived Columns and Modeling Features Notes

## Purpose

This document summarizes the project-specific target variables, modeling frames, and feature engineering used in the final capstone notebooks. It is not intended to replace the client-provided variable descriptions.

## Task A: Current GSTD Classification

Task A predicts the current growth stage, GSTD. The final Task A models compare broader benchmark feature sets against practical client-feasible feature sets.

- The target variable is GSTD.
- Client-feasible modeling uses the agreed practical variables available to the client workflow.
- LAID is included in the strongest feasible Task A model as the available LAI-equivalent measure.
- Time-series features are used in some Task A models to summarize recent past conditions before the prediction day.

## Task B Target: y_enter5_7

Task B predicts whether a crop observation will enter GSTD 5 within the next 7 days. The binary target created for this purpose is named y_enter5_7.

- y_enter5_7 = 1 means the observation enters GSTD 5 within the next 7 days.
- y_enter5_7 = 0 means the observation does not enter GSTD 5 within the next 7 days.
- The target is created from the ordered growth-stage sequence within each simulation run.

## Task B At-Risk Framing

The final Task B modeling frame focuses on observations where the 7-day GSTD 5 alert is meaningful. In the final dataset, positive 7-day GSTD 5 transitions arise from current GSTD 1 and GSTD 3. GSTD 3 is effectively deterministic under this 7-day target structure, so the final early-warning model focuses on Stage 1 observations.

- The final predictive Task B model is framed as a Stage 1 early-warning model.
- This avoids overstating performance by including obvious non-risk rows.
- This framing better reflects the operational alerting problem: among current GSTD 1 observations, which ones will enter GSTD 5 within the next 7 days?

## Time-Series Features

Some final models use past-looking time-series features. These features summarize recent conditions before the prediction date and are designed to avoid using future information.

- Lag features summarize recent prior values.
- Rolling summary features capture short-term patterns across recent days.
- Difference/change features capture recent movement in weather, crop, or agronomic variables.
- The final Task B recommendation uses client-feasible variables with time-series engineering.

## LAID

LAID was included as the available LAI-equivalent variable. It was tested because of its connection to leaf area and because the client specifically asked whether LAI-related information should be used.

- For Task A, Client-6 + LAID + time-series features was the strongest feasible version.
- For Task B, adding LAID on top of time-series features produced only a very small performance gain.
- For the final Task B recommendation, Client-Feasible TS LightGBM is the core model, and Client-6 + LAID + TS LightGBM is an optional extension if LAID is reliably available.

## Validation Columns and Splits

Several dataset columns are used to create honest validation splits rather than random row-level splits.

- RUNNO is used for grouped validation so observations from the same simulation run are not split across train and test folds.
- Year is used for the forward-time holdout to test how models perform on future years.
- Treatment identifiers are used for treatment-holdout robustness checks.

## Final Task B Operating Threshold

The final Task B LightGBM operating threshold is selected from forward-time holdout predictions. In the finalized run, the recommended Client-Feasible TS LightGBM model had a Best F1 operating threshold of 0.9984.

- At this threshold, the model caught 335 upcoming GSTD 5 transitions.
- The model missed 69 transitions.
- The model generated 62 false alerts.
- The alert rate was about 30.3 percent across the future-year holdout observations.

## Important Notes

- The final dataset is DSSAT/simulation-derived, so real-world validation is still needed before operational deployment.
- The final notebooks compute the derived target and feature sets directly from final_validated_dataset.csv.
- The exact engineered feature names may differ depending on the notebook section, but they are created from past-looking values only.
- The final model should be interpreted as a decision-support alerting tool, not an automatic agronomic decision system.